19/01/26

EE–556 NUCLEAR REACTOR CONTROL

Lecture–2: Reactor Physics Preliminaries

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Course Schedule

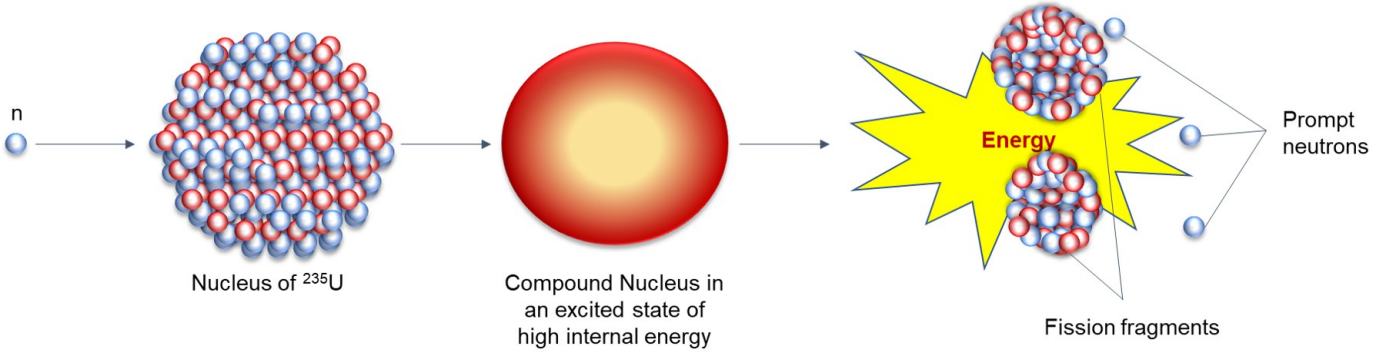
- **Course Registration:** Completed. Class strength is 65.
- **Course Slot:** No slot
- **Lecture Schedule:**
  1. 2 contact hours (2 lectures $\times$ 50 minutes) per week on:
     Jan. 19M and 24S/28W (in lieu of 26th),
     Feb. 2M, 9M, 18W(in lieu of 16th) and 23M(online),
     March 11W (online), 16M, 23M, 30M,
     April 6M, 13M, 20M (Reserved dates).
     Online lectures – in case of classroom unavailability.
- **Course Completion:** April 6, 2026/April 13, 2026
- **Lecture slides:** will be uploaded on Moodle after the lecture.

Scheme of Evaluation and Attendance Requirement

- **Assignment 1 (10–15 %),** Feb. 9, 2026 (tentatively)
- **Assignment 2 (10–15 %),** Mar. 12, 2026 (tentatively)
- **End-sem Exam (70–80 %),** on April 18/25 (Saturday)/ any other day during April 18–25, 2026.
- **Passing Marks:**
  1. For Regular Students: 30 %. Relative grading will be done only for students obtaining $\geq$ 30 %.
  2. For Pass/Fail Students: 45%
- **Attendance Requirement:** 80 %
  (Attendance will be taken 2 times in each lecture.)

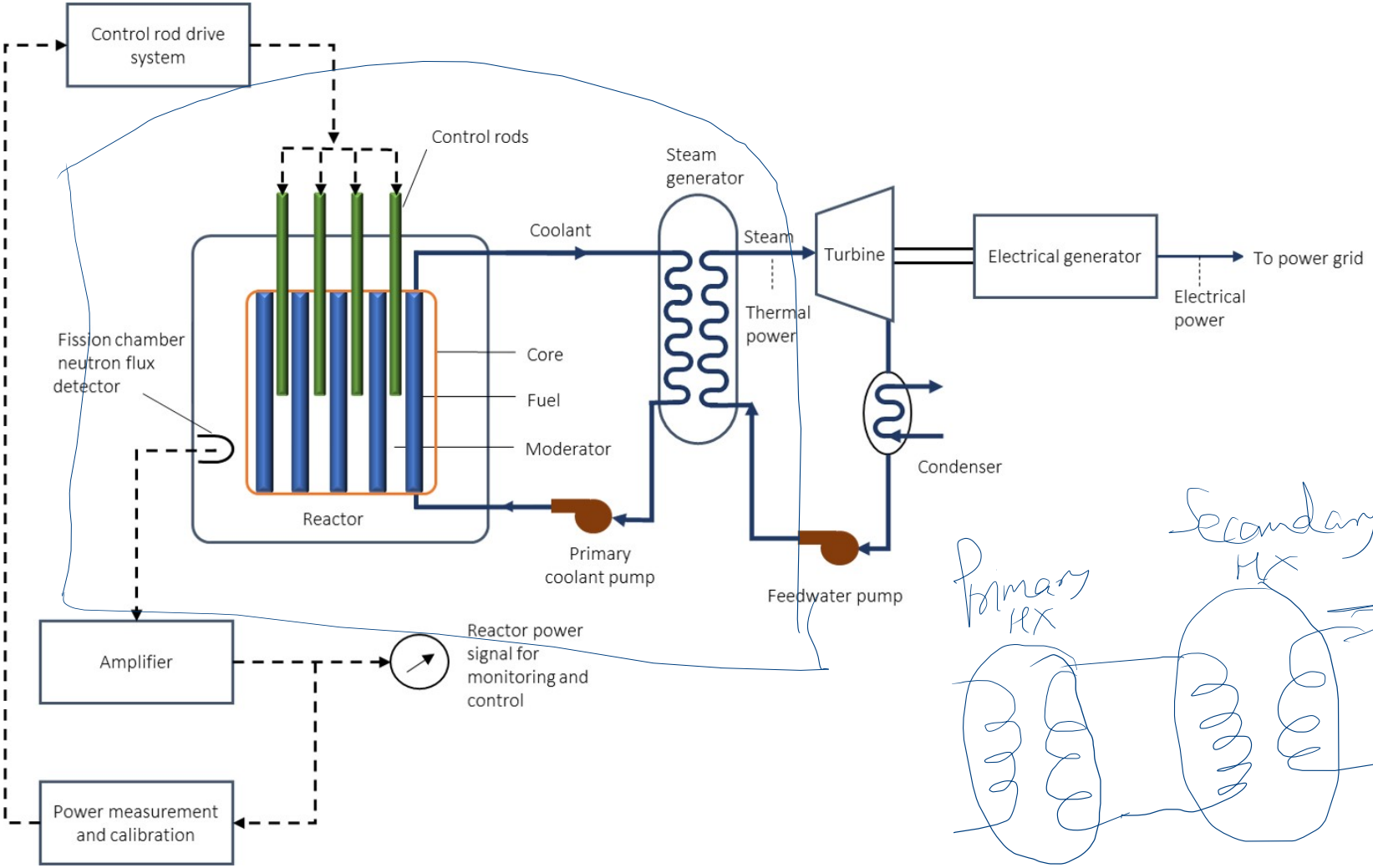
Introduction: A Nuclear Reactor

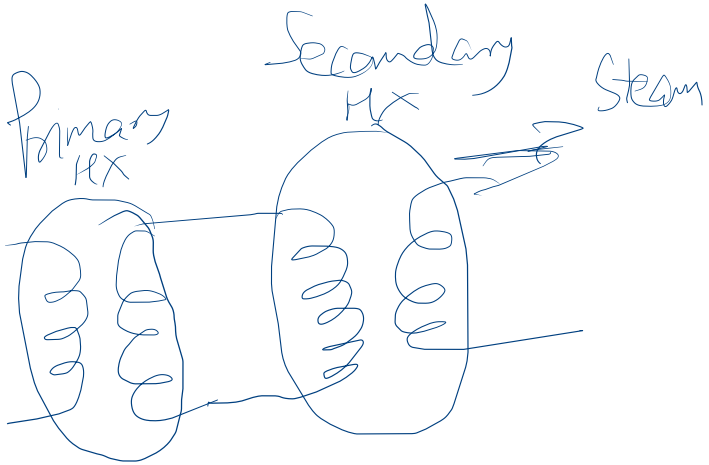
Nuclear Reactor: A nuclear reactor is a device in which controlled fission chain reactions take place.



Typically in a nuclear reactor, neutrons are absorbed in the fuel (e.g., Uranium-235) nuclei, which undergo fission resulting into the production of fission fragments, neutrons, other radiations and release of approximately 200 MeV energy per fission.

NPP Schematic





Introduction (Contd.): Reactor Types

- Classification Based on Neutron Energy
  1. **Fast Reactors**, in which fission reactions are maintained by fast neutrons.
  2. **Thermal Reactors**, in which fission reactions are maintained by thermal neutrons.
- In India:

  Fast breeder test reactor at Kalpakkam, is the only **fast** reactor operational now.

  Prototype fast breeder reactor (PFBR) is **power fast** reactor under commissioning at Kalpakkam.

  All other reactors are thermal reactors.

NPCIL

NTPC

Introduction (Contd.): Reactor Types

- Classification Based on the Coolant type

Thermal reactors

Coolant + Moderator

1. Light Water Cooled Reactors
   - **BWR** Boiling Water Reactor
   - **PWR** Pressurized " "
2. Heavy Water Cooled Reactors
   - **PHWR** Pressurized Heavy Water Reactor
   - **AHWR** Advanced "
3. Gas Cooled Reactors
4. Liquid Metal Cooled Reactors

Fast Reactor

^1_1H

H_2O at low pressure

" at relatively higher pressure

D_2O

^2_1H

AHWR H_2O

D_2O

Enriched U

In PHWR

Coolant D_2O

Moderator D_2O

Fuel Natural U

Introduction (Contd.): Main Components of a Nuclear Power

Natural U — ^{238}U & ^{235}U
>98% 0.7%

- Core, consisting of
  1. Fuel $^{235}_{92}U$,
  2. Moderator
  3. Coolant
  4. Control rods/elements
- Reflector
- Shield
- Neutron flux sensors, can be **incore** or **excore**

Enriched fuel
⇓
Content of ^{235}U is higher than 0.7%

Moderator

H_2O (Light water)

D_2O (Heavy ")

Graphite

Beryllium

Coolant

H_2O

D_2O

Na

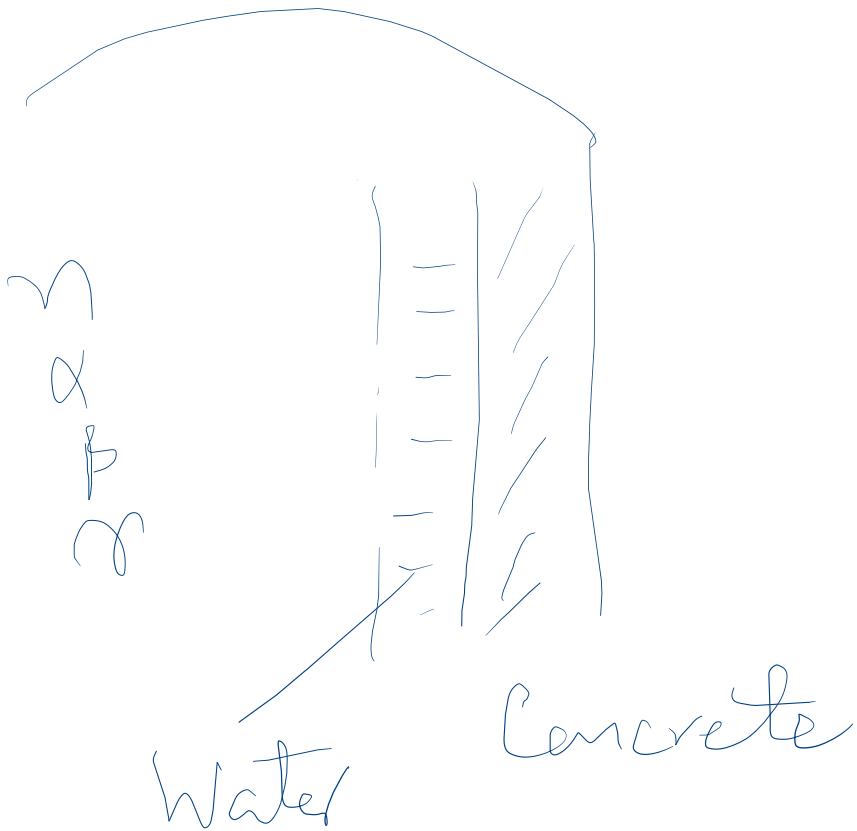
NaK

Pb-Bi Eutectic

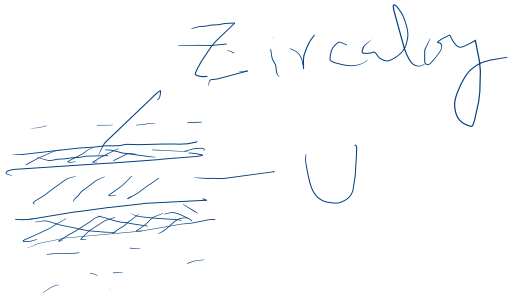
Control Boron / Steel – in fast Reactors

Cd in Thermal reactors

Shielding



Clad



Nuclear Reactor Applications

1. Electrical Energy Generation
2. Process Heat Generation
3. Desalination
4. Isotope Production
5. Neutron Source
6. Hydrogen Production

Emerging Trend: Small Modular Reactors

1. Current Trend: Large Reactors
2. Earlier Generation Designs: Small and Medium–sized reactors, typically from approx. 100 MWe to 300 MWe
   - Small number of these facilities continue to be operational today.
   - They evolved as a result of efforts to experiment with different cooling technologies, fuel types and operating configurations.
   - They were not designed at that time for modular fabrication or construction.

Emerging Trend: Small Modular Reactors

1. SMRs are small reactors designed to produce upto 300 MWe power.
2. SMRs are expected to capture in the short-to medium-term, a significant share of the market in the nuclear sector.
3. Designed so as to reduce the total capital costs of nuclear power plants and to provide power to small grid systems.
4. Factory fabricated and transported to operation–site.
5. SMRs have the specific characteristics to match the spiralling energy demand by adding incremental capacity with moderate financial commitment.

Course Objective, Contents, and Outcome

- **Course objective:** To apply control system modeling, analysis and design principles to nuclear reactor, visulised as as a controlled system
- **Contents**
  1. Nuclear fission, Neutron Diffusion equation
  2. Derivation of the Point kinetics model of nuclear reactor and its solution to standard inputs
  3. Study of nuclear reactor as a control system element - investigation of stability, controllability and observability
  4. Reactor instrumentation; Reactor Control Methods
- **Course outcome:** The learner will be familiar with basic principles and methods of reactor control.

Recapitulation of the last lecture

- Notations
  1. ${}^{A}_{Z}\mathrm{X}$ = a specific nucleus with the chemical symbol X.
  2. Nuclides: Various species of nuclei
  3. Isotopes: nuclei having the same Z but different A
  4. ${}^{A}_{Z}\mathrm{X}^{*}$ = nucleus in excited state
  5. ${}^{A}_{Z}{}^{m}\mathrm{X}$ = isomer, i.e., long lived excited state of nucleus X
- Mass defect $\Delta m = [\underbrace{Z\, m_P}_{\text{protons}} + \underbrace{(A - Z)\, m_N}_{\text{neutrons}}] - m_{nucleus}$
- BE $= \Delta m \cdot c^2$

Recapitulation (contd.): Nuclear Reactions

1. Spontaneous disintegration of nuclei– depends purely upon the properties of an individual nucleus.
2. Reactions resulting from collision between nuclei and/or nuclear particles.

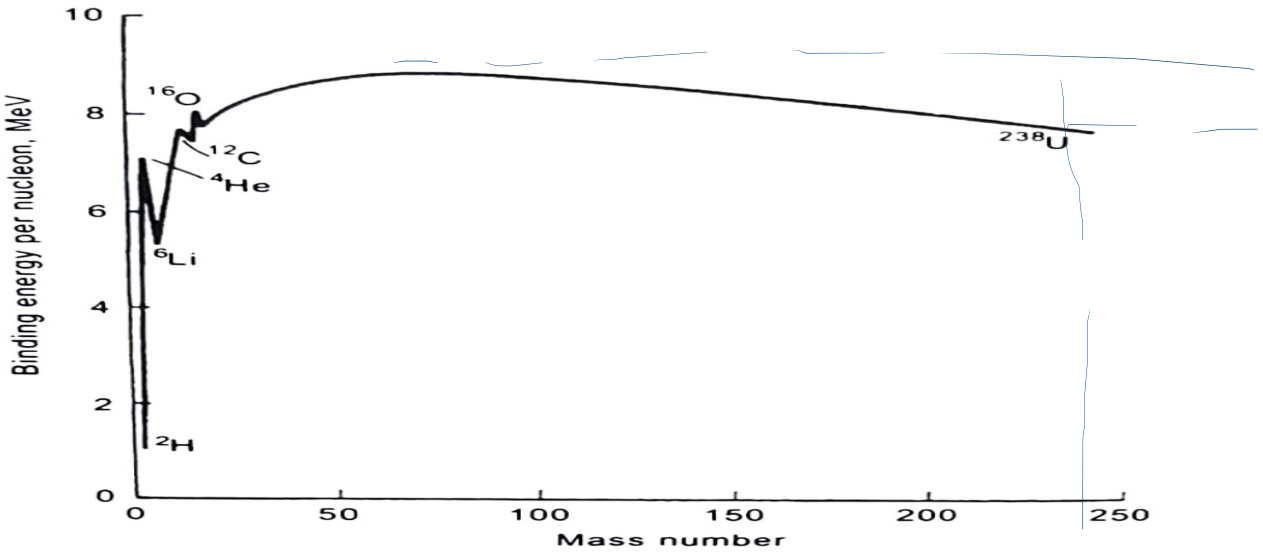
   These depend upon

   1. properties of the colliding particles/nuclei, and
   2. relative velocities with which collision occurs.

   Among these are

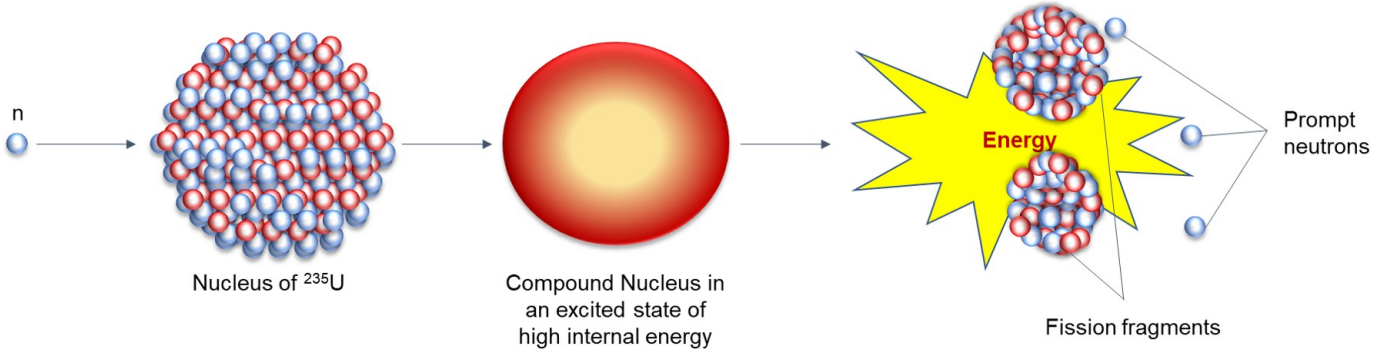
   1. Neutron–induced nuclear fission
   2. Radiative capture
   3. Scattering (Elastic and Inelastic)
3. Other, e.g., (n,α), (α,n), (γ,n), and (γ,e) reactions.

Neutron–Induced Fission



- When a neutron is bombarded on a target nucleus (A,Z), to form the compound nucleus (A+1,Z), then
- the BE per nucleon of the *compound* nucleus $<$ BE per nucleon of the *target* nucleus.
- This reduction may be sufficient that the compound nucleus will undergo fission even if the incident neutron has low energy.

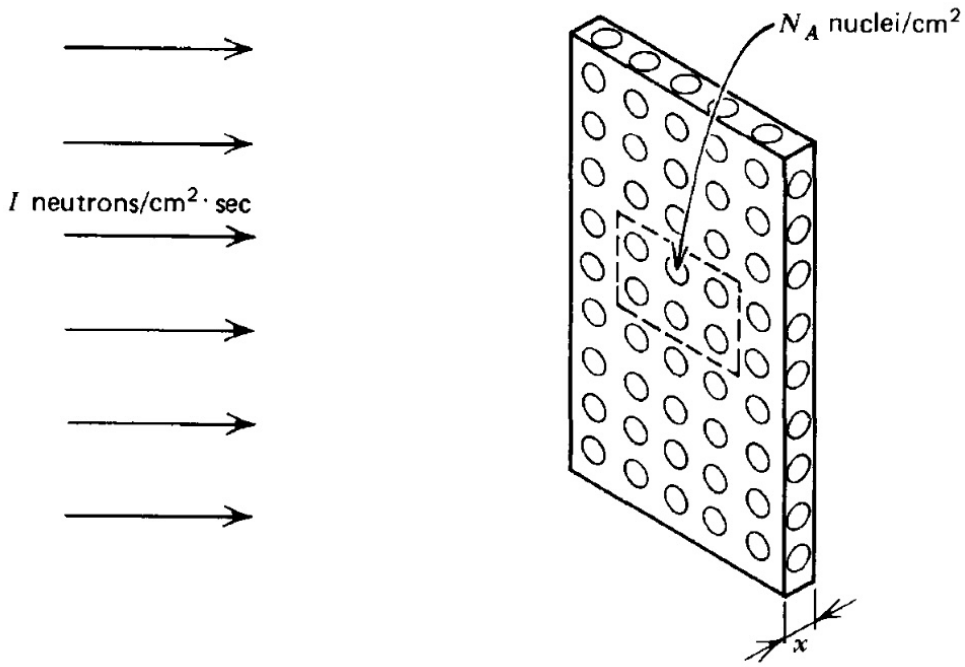
Neutron–Induced Fission (Contd.)

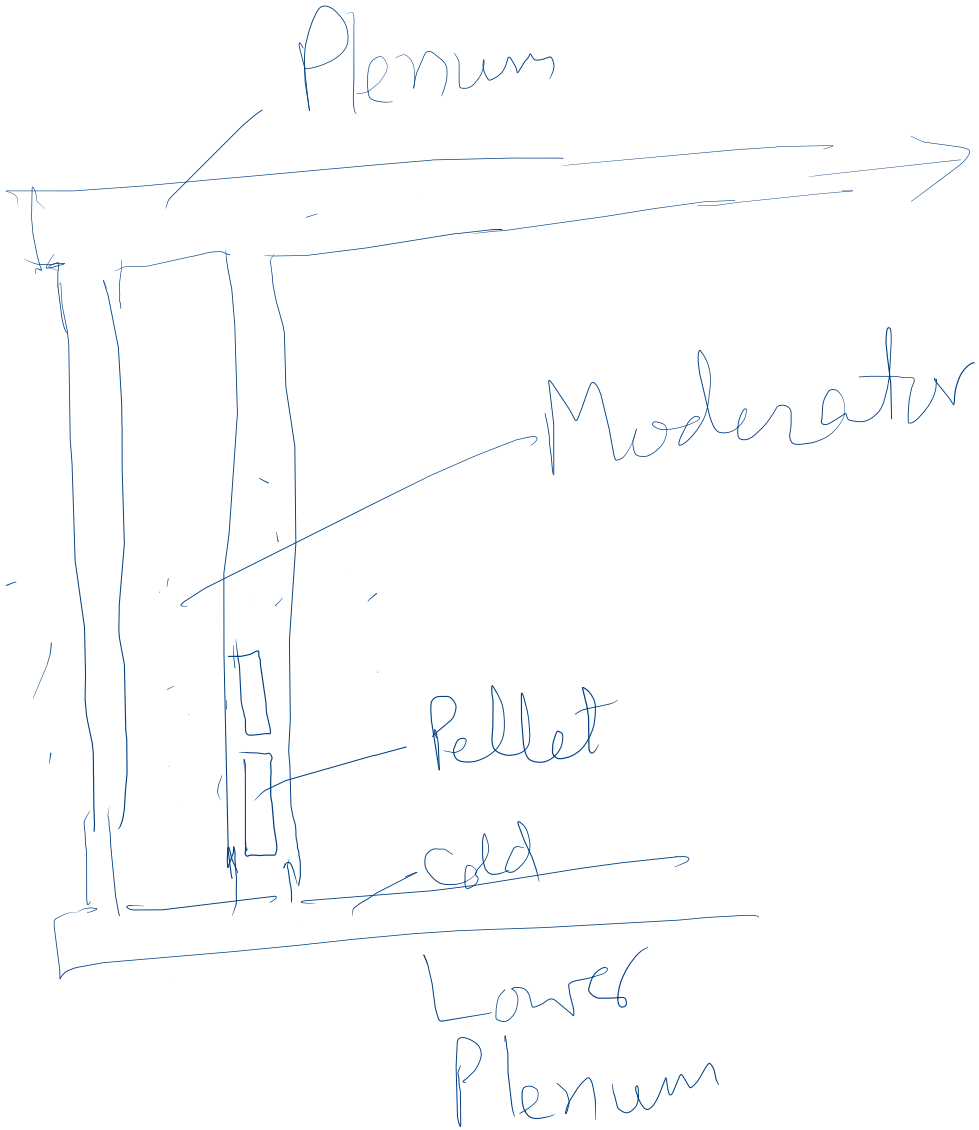


- Fissile nuclides (e.g., ${}^{233}_{92}U$, ${}^{235}_{92}U$, ${}^{239}_{94}Pu$, ${}^{241}_{94}Pu$) have high probability of undergoing fission on bombardment by low energy neutrons.
- Fissionable nuclides (e.g., ${}^{232}_{90}Th$, ${}^{238}_{92}U$, ${}^{240}_{94}Pu$) will undergo fission on bombardment by high energy neutrons

Nuclear Cross sections

- The probability that a neutron–nucleus interaction will occur, is characterized by **cross section**.
- Consider a beam of neutrons, all travelling in the same direction with the same speed, incident normally and uniformly upon the face of a target material.



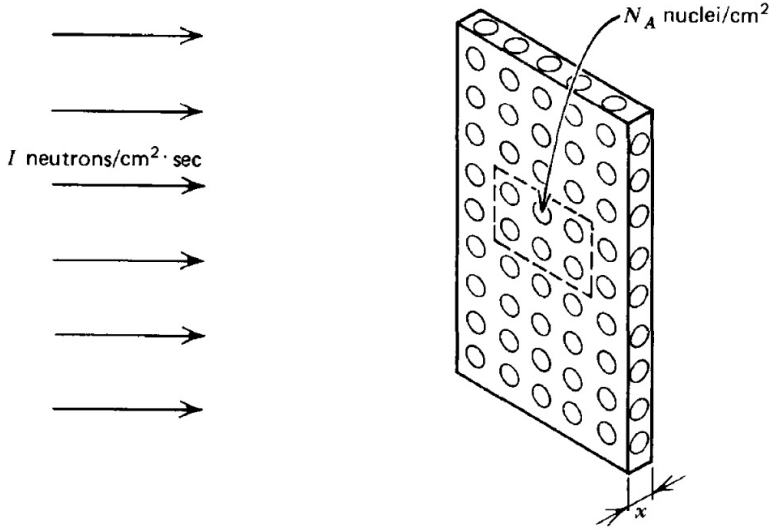

Plenum
Moderator
Pellet
Cold
Lower
Plenum

Nuclear Cross sections

- The target is assumed to be thin so that the nuclei are not schielded by other nuclei.
- Then the rate of neutron nuclear interaction,

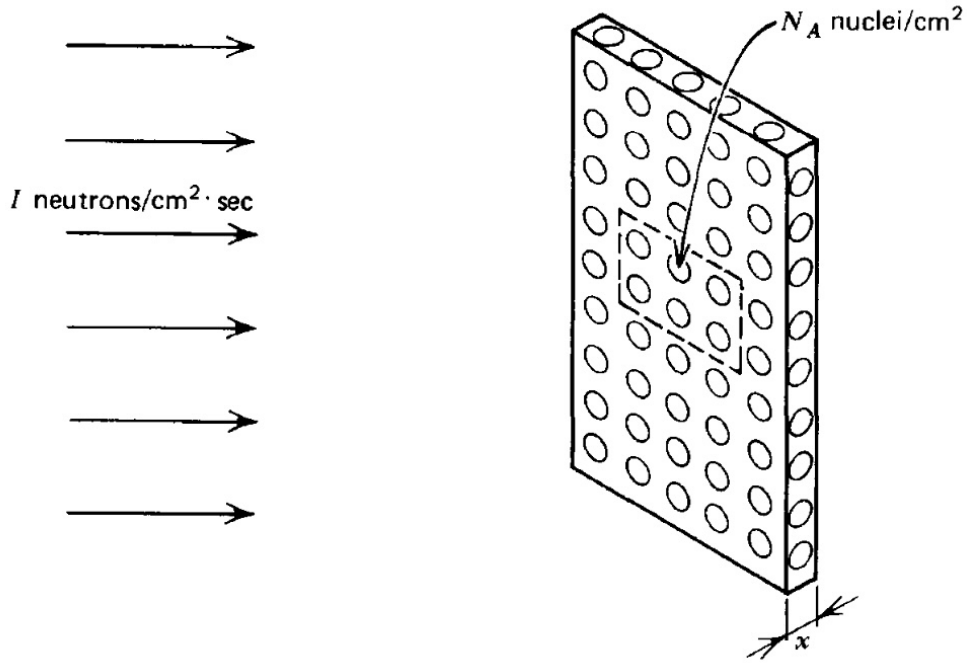
$$R \propto I\, N_A \quad \text{or,} \quad R = \sigma\, I\, N_A,$$

where σ is called the microscopic cross section. Its unit is barn. 1 barn $= 10^{-24}$ cm^2.

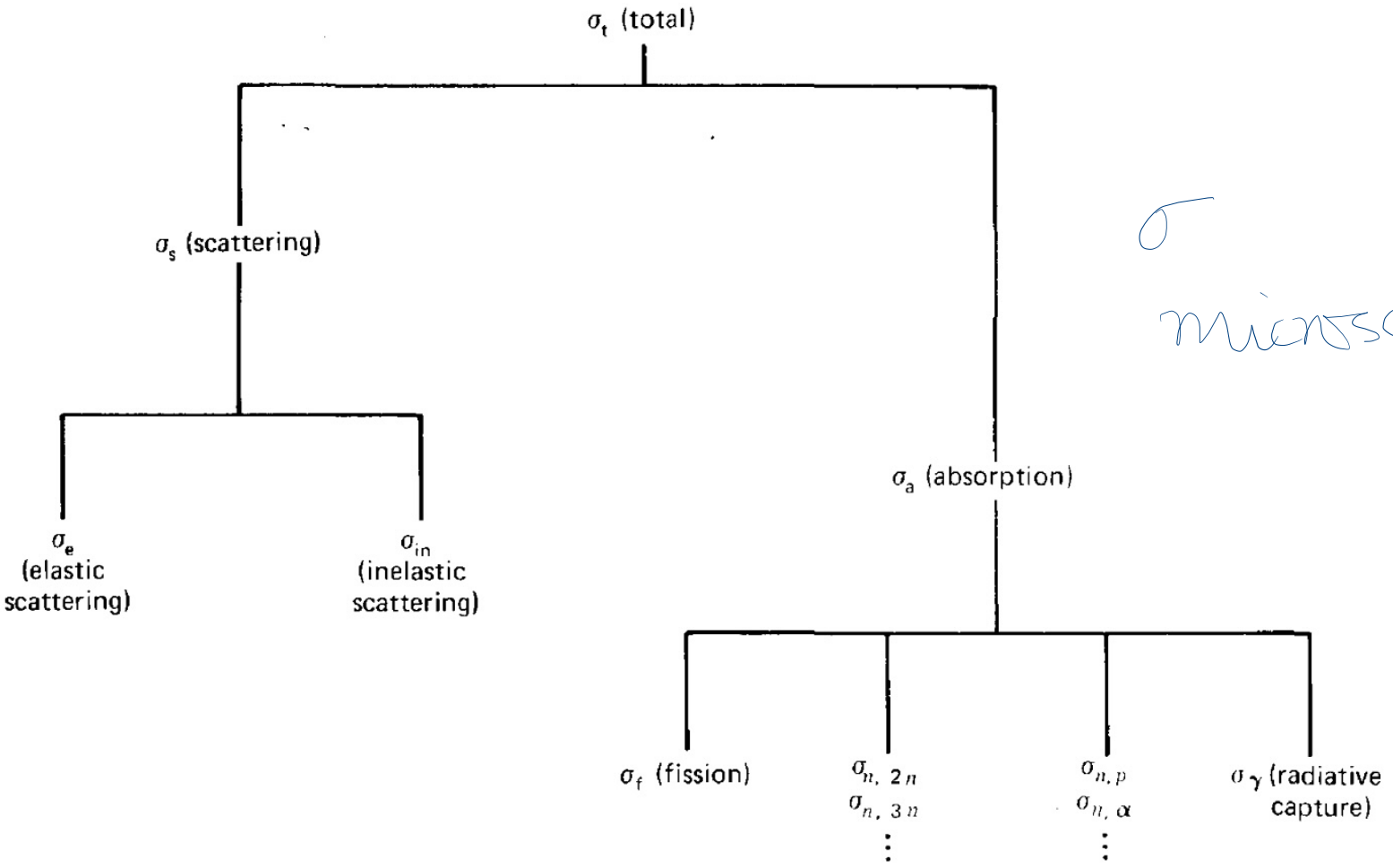


Nuclear Cross sections

- Fission cross section, σ_f is the measure of the probability that a neutron and a nucleus interact to form a compound nucleus which then undergoes fission.
- Other reaction cross sections can be defined in a similar manner.

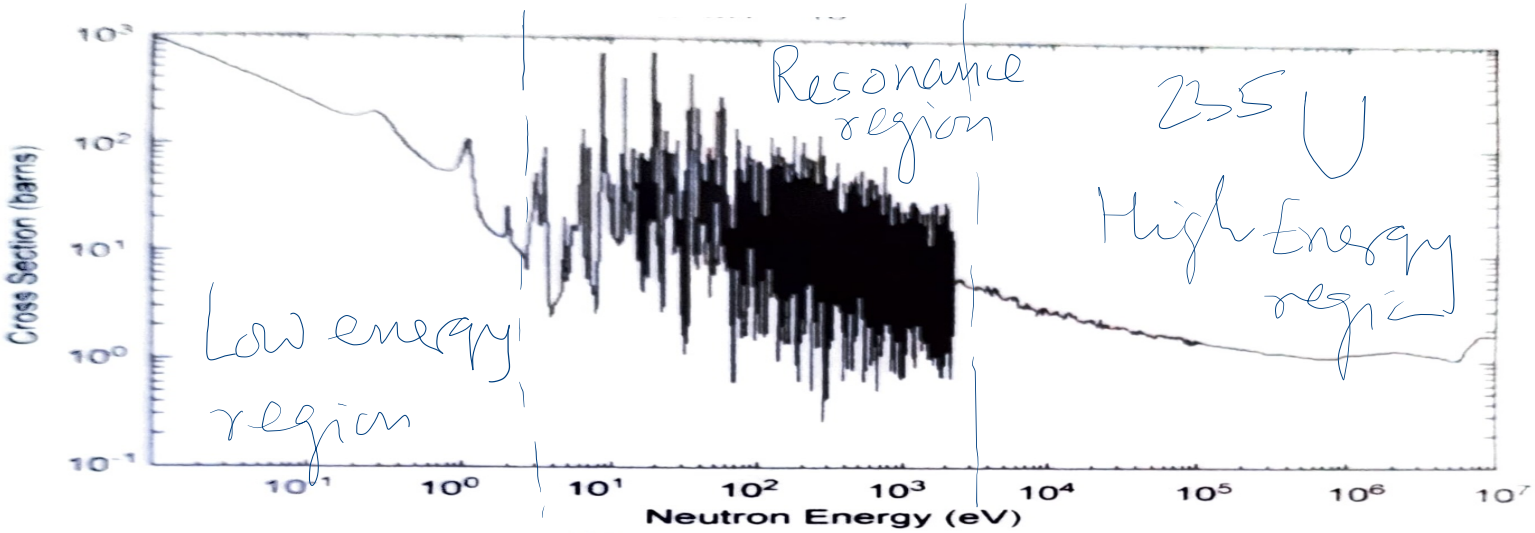


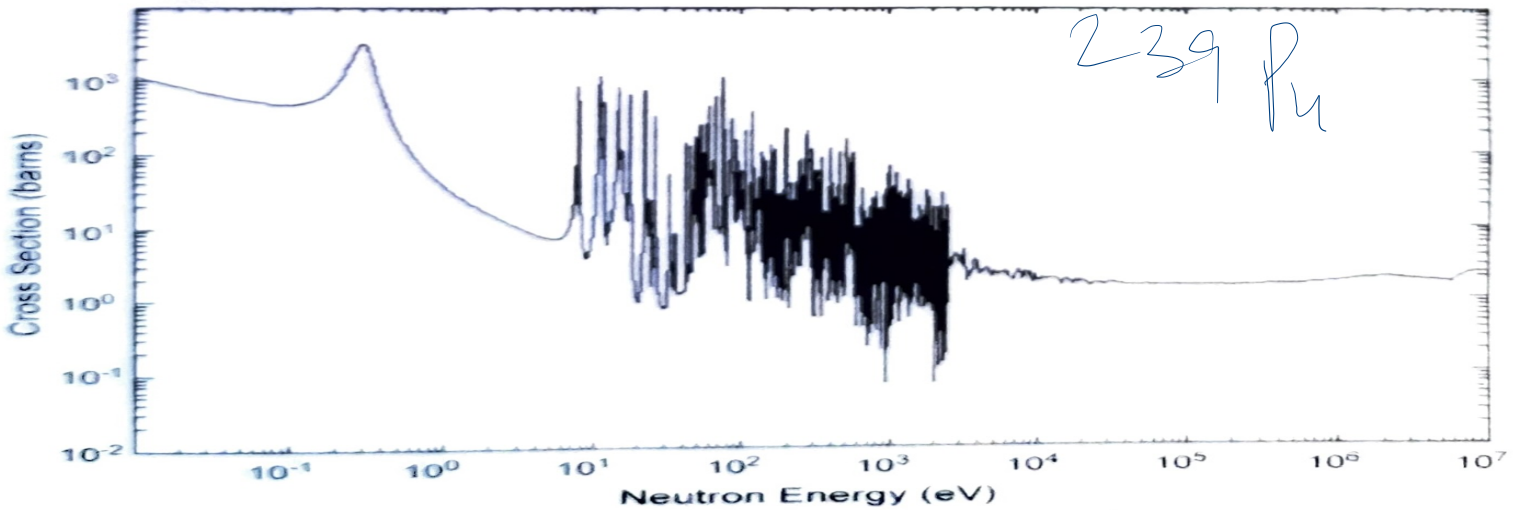
Nuclear Cross sections

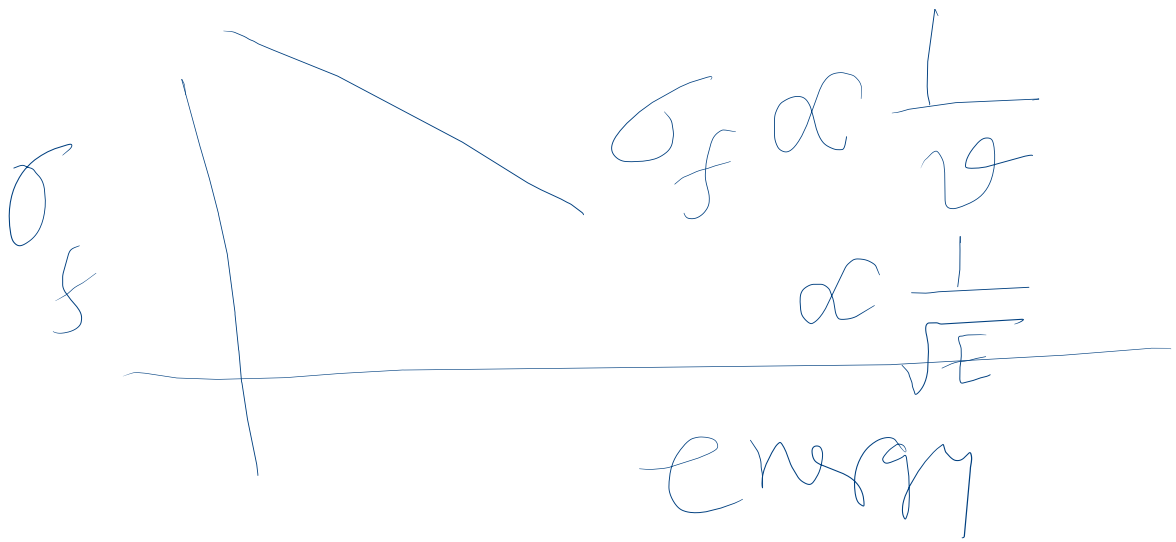


σ microscopic

Fission Crosssection energy dependence for U & Pu






σ_f
$\sigma_f \propto \frac{1}{v}$
$\propto \frac{1}{\sqrt{E}}$
energy

Macroscopic Crosssection

- The product of number density, N and the microscopic crosssection, σ is called the macroscopic crosssection, denoted by Σ.

  $\frac{\text{number}}{cm^3}$ cm^{-2}

- That is, $\Sigma = N\sigma$.
- Σ has dimensions of inverse length, cm^{-1}.
- It represents the probability per unit path length in a medium that the neutron will undergo a reaction.
- $v\Sigma$ represents the frequency with which the reactions occur.

  $\frac{cm}{s} \times \frac{1}{cm} = \frac{1}{s} = s^{-1}$

- $(v\Sigma)^{-1}$ represents the mean time between neutron interactions.

Neutron Nuclear Crosssection for mixture

per cm^3

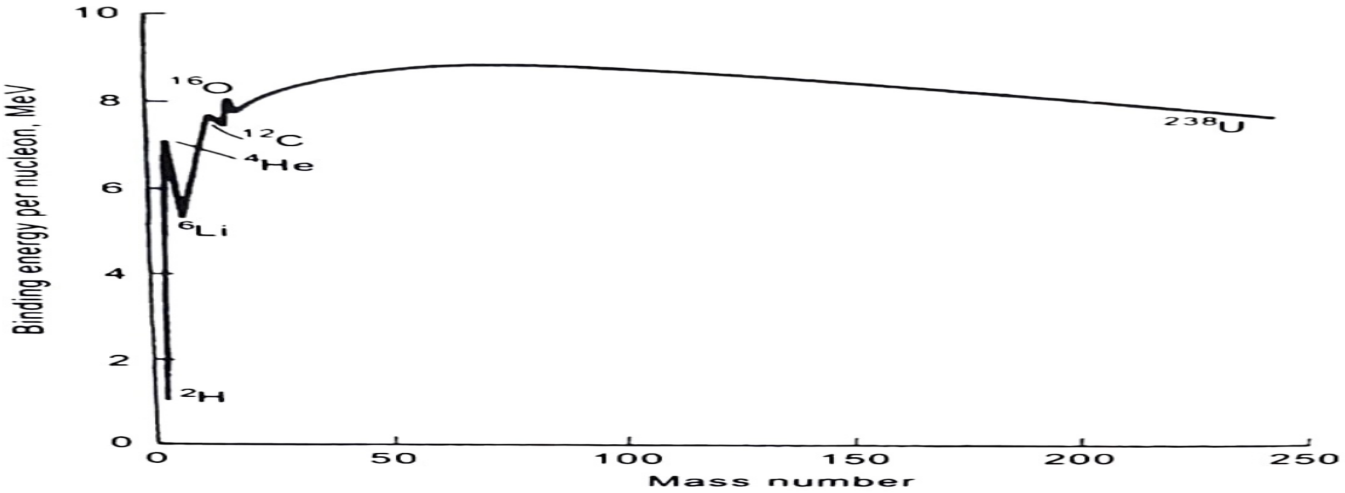
Mixture N_1 N_2 N_3 ...

nuclei of type 2 type 2 type 3

microscopic crosssections σ_1 σ_2 σ_3 ...

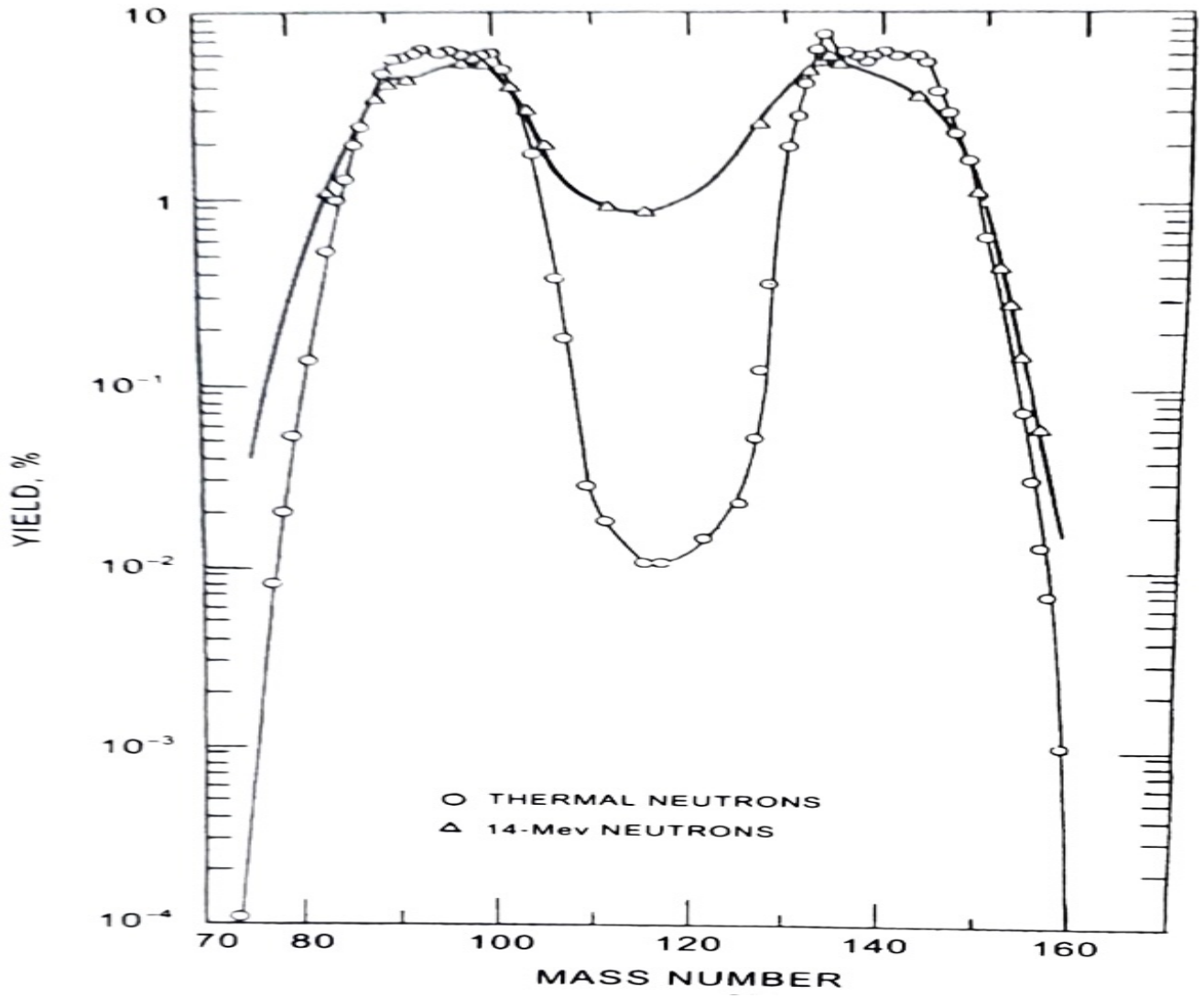
$$\bar{\sigma} = \frac{\sum N_i \sigma_i}{\sum N_i}$$

Threshold External Energy for Fission Reaction



- The probability of a nuclide undergoing fission can become large if sufficient amount of external energy is supplied to excite the nucleus.
- For nuclides with $Z > 90$, the threshold energy is
  - about 4–6 MeV for nuclides with even A.
  - much lower for nuclides with odd A.

Products of the Fission Reaction–Fission fragments



Products of the Fission Reaction–neutrons

Prompt and Delayed Neutrons

- Most of the fission neutrons are released almost instantaneously (within 10^{-14} s).

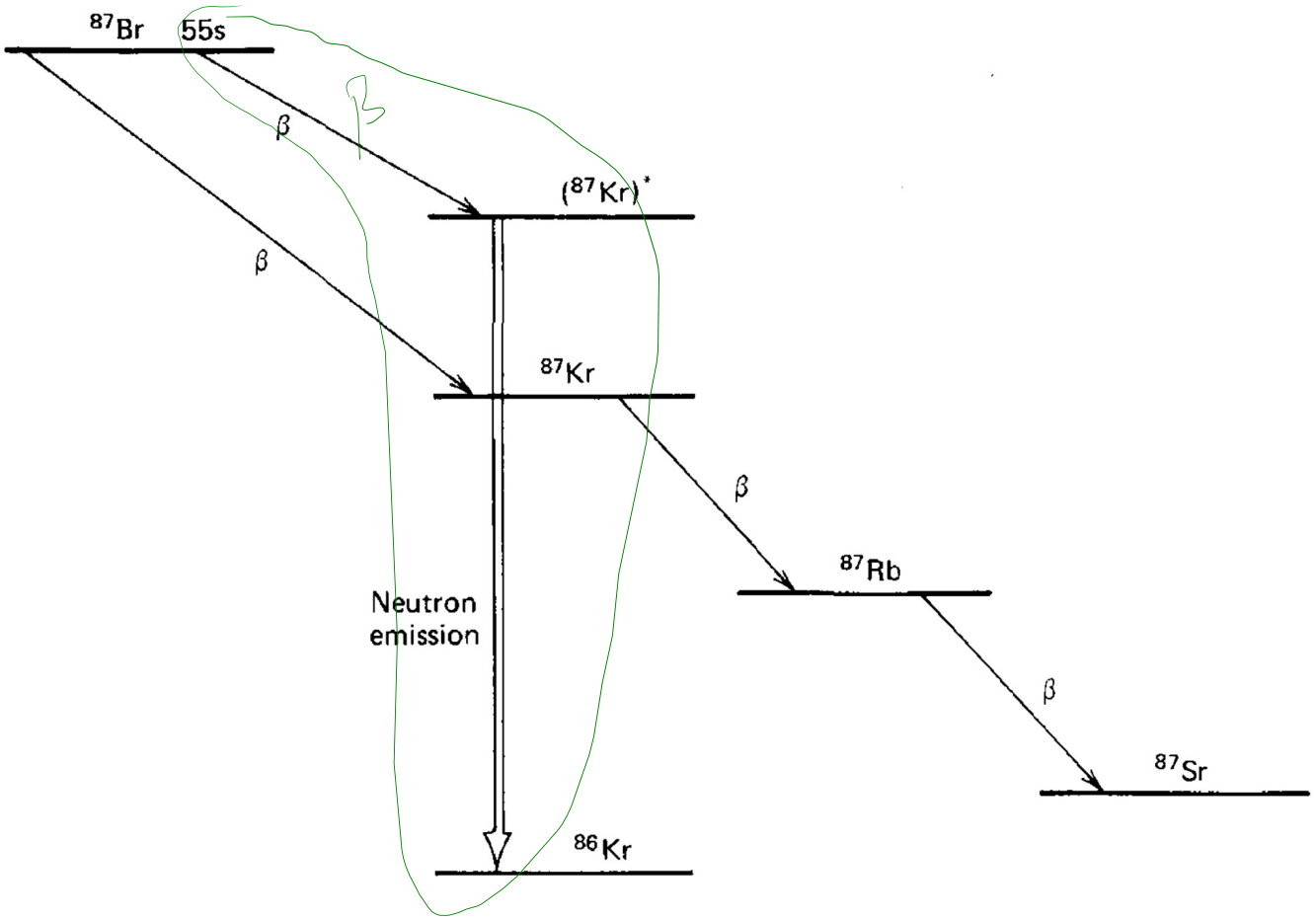
  These are referred to as *prompt* neutrons.

- A very few neutrons (less than 1%) appear with an appreciable time delay from the fission event.

  These neutrons are referred to as *delayed* neutrons.

Products of the Fission Reaction–neutrons

Delayed Neutrons production mechanism: 55 s group



Products of the Fission Reaction–Delayed neutrons

Delayed Neutrons production mechanism: 23 s group

$$^{137}_{53}\mathrm{I} \xrightarrow{23\,s} (^{137}_{54}\mathrm{Xe})^{*} \longrightarrow {}^{1}_{0}\mathrm{n} + {}^{136}_{54}\mathrm{Xe}$$

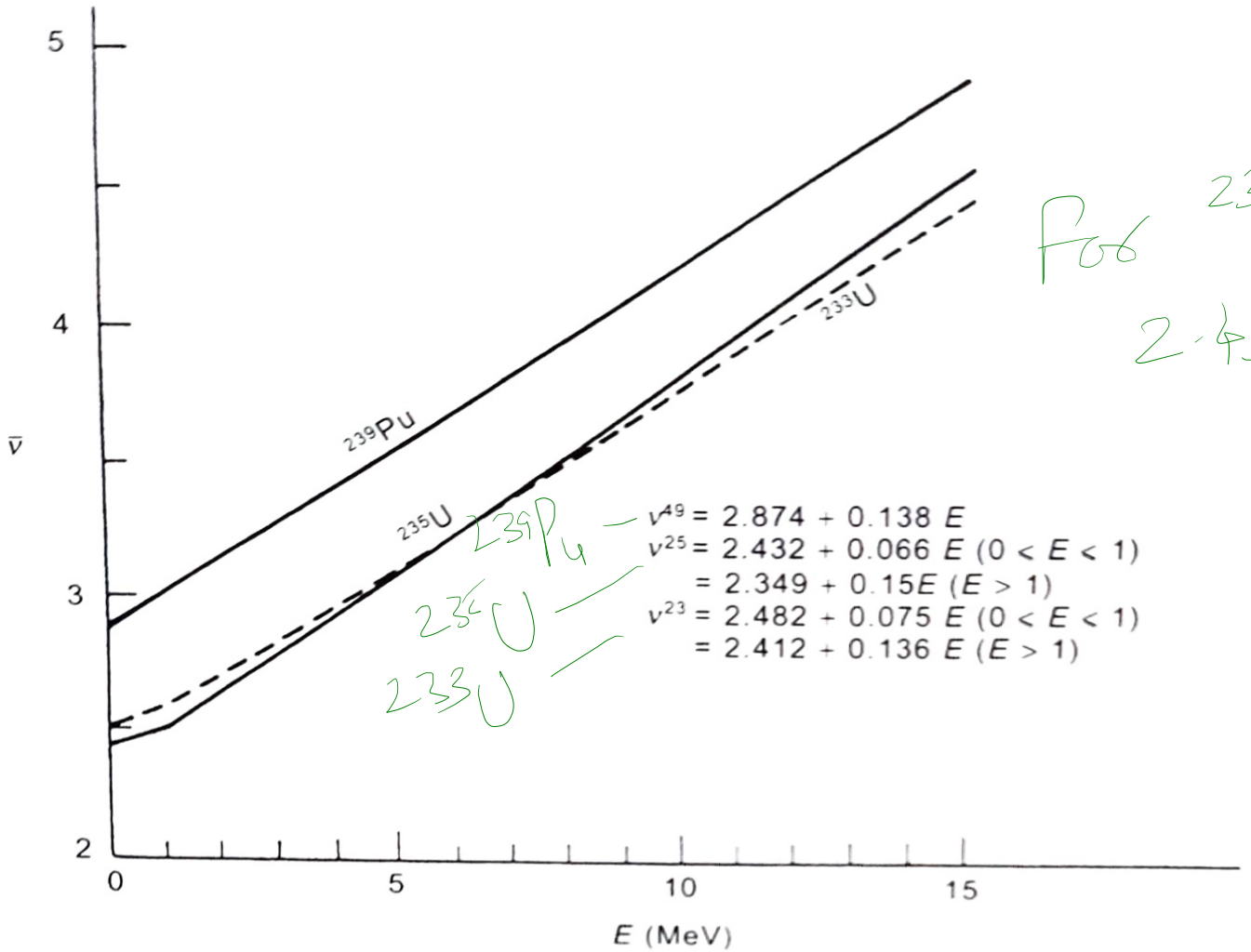
Products of the Fission Reaction–Delayed neutrons

Delayed Neutron groups

TABLE 2.10. Characteristics of Delayed Fission Neutrons in Thermal Fission

Approximate Half-Life (seconds)	Number of Fission Neutrons Delayed per Fission			Energy (MeV)
	U-233	U-235	Pu-239	
55	5.7×10^{-4}	5.2×10^{-4}	2.1×10^{-4}	0.25
23	19.7	34.6	18.2	0.46
6.2	16.6	31.0	12.9	0.41
2.3	18.4	62.4	19.9	0.45
0.61	3.4	18.2	5.2	0.41
0.23	2.2	6.6	2.7	—
Total delayed	0.0066	0.0158	0.0061	
Total fission neutrons	2.49	2.42	2.93	
Fraction delayed	0.0026	0.0065	0.0020	

Products of the Fission Reaction–neutrons



For ^{235}U 2.43 neutrons per fission

Fig. 1.10 Average number of neutrons emitted per fission.

Fission spectrum for thermal neutron fission of ^{235}U

Prompt Neutrons

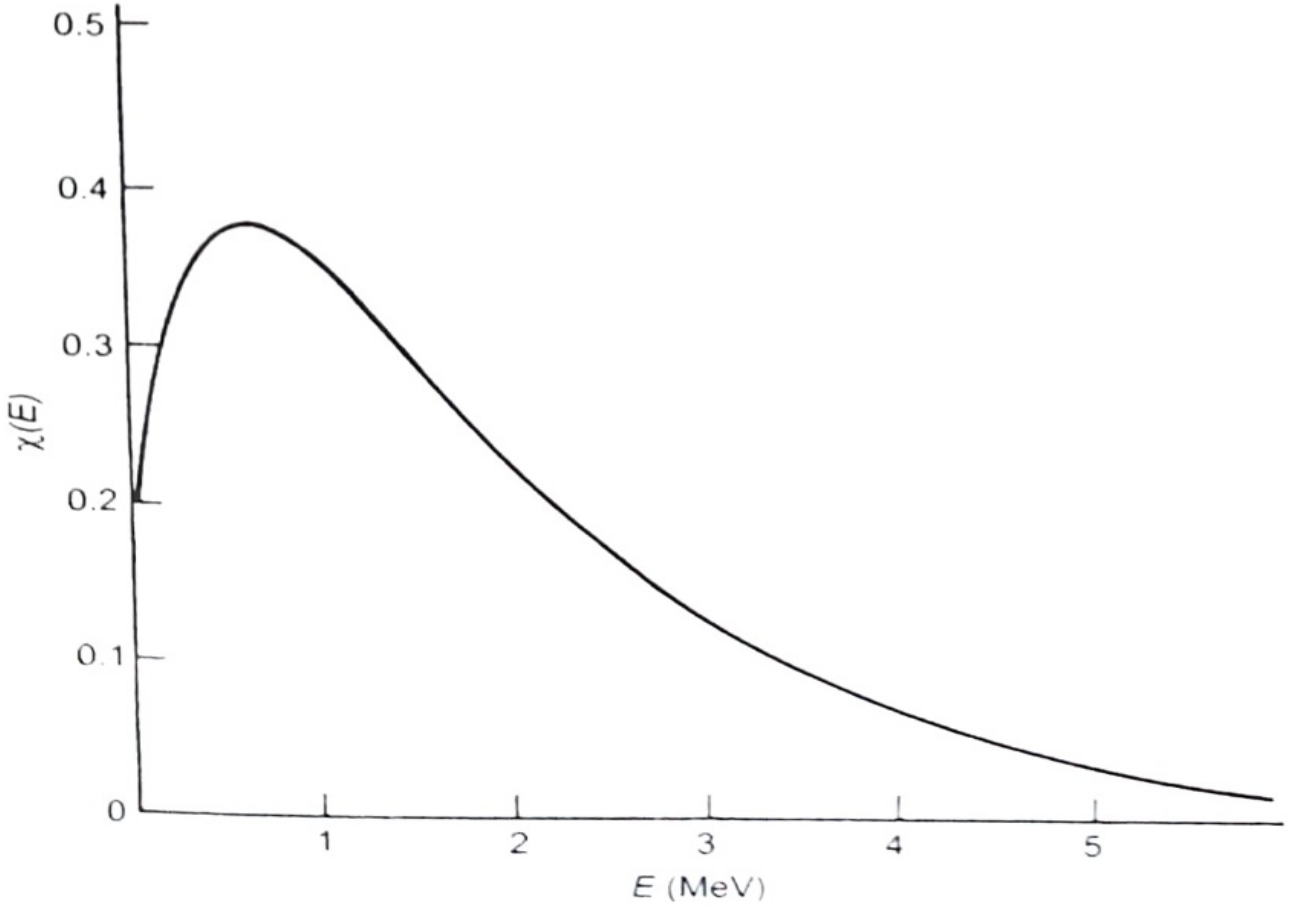
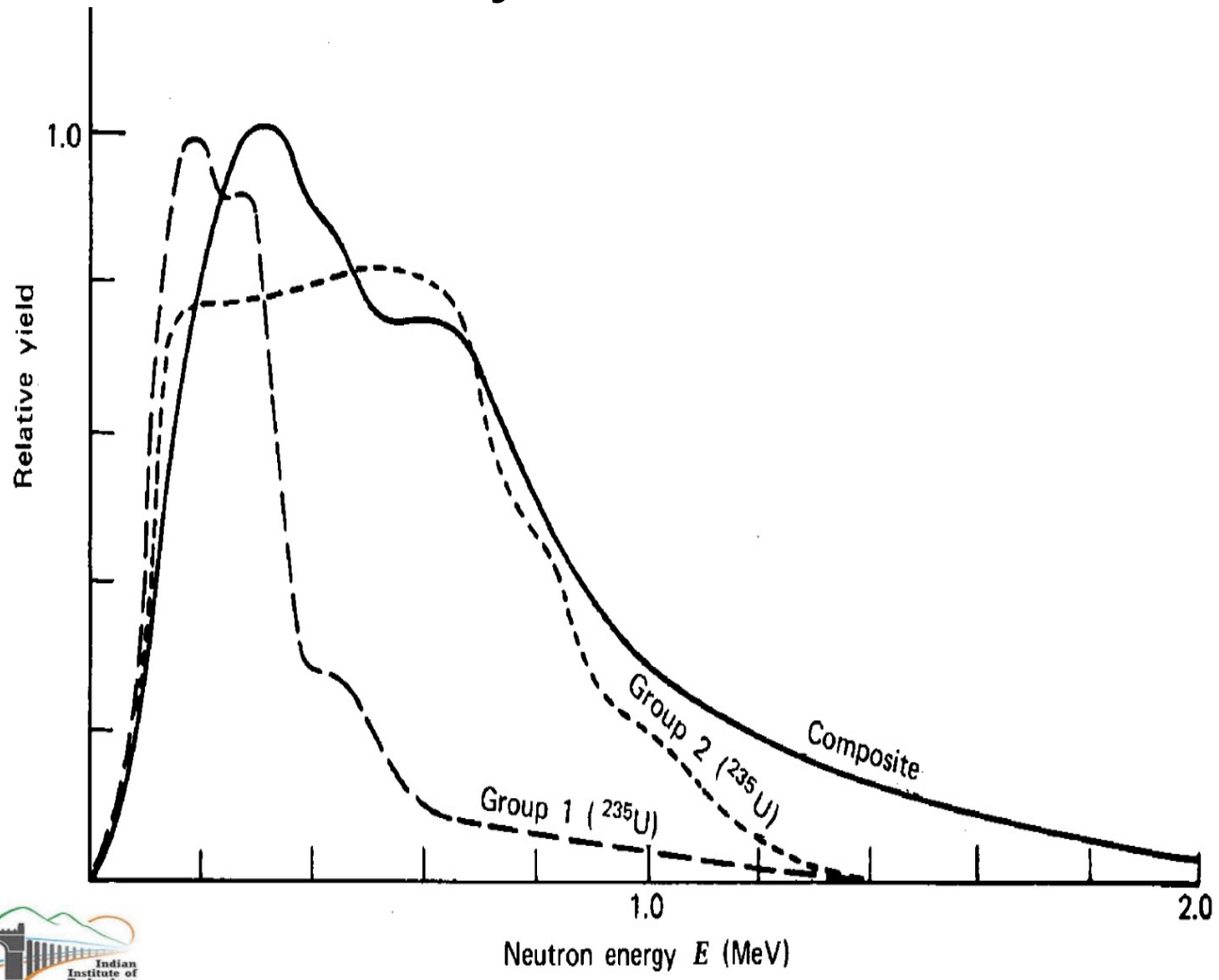


Fig. 1.11 Fission spectrum for thermal neutron-induced fission $^{235}U_{92}$. (From Ref. 12; used with permission of Wiley.)

Fission spectrum for thermal neutron fission of ^{235}U

Delayed Neutrons



Energy Released in Fission Reaction

Table 1.1 $^{235}U_{92}$ Fission Energy Release

Form	Energy (MeV)	Range
Kinetic energy fission products	168	< mm
Kinetic energy prompt gammas	7	10–100 cm
Kinetic energy prompt neutrons	5	10–100 cm
Kinetic energy capture gammas	7	10–100 cm
Decay of fission products		
Kinetic energy electrons	8	~mm
Kinetic energy neutrinos	12	∞